

NPTEL Course on "Computer System Optimization for ML"

Lecture #29

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Agenda

- **Paged-attention (contd.)**

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Chunked Preallocation - Drawbacks

- KV cache stored in contiguous memory
- Static allocation based on max request needs
- Ignores the fact that memory is needed incrementally and not all at once

- **Internal Fragmentation due to overprovisioning for max tokens**
- **External Fragmentation due to buddy memory allocators**
- **Underutilized reserved space due to static allocation**

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vLLM : Paged Attention

Key Insight :

We need dynamic memory allocation for KV caches

– Allocate on demand as a request grows

Inspired by OS **virtual memory** and **paging**!

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vLLM : Paged Attention

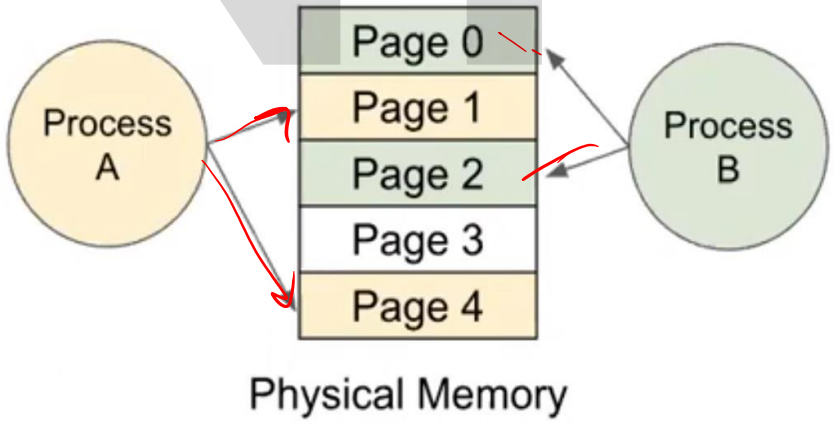
Key Insight :

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Inspited by OS **virtual memory** and **paging**!

Memory management in OS



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vLLM : Paged Attention

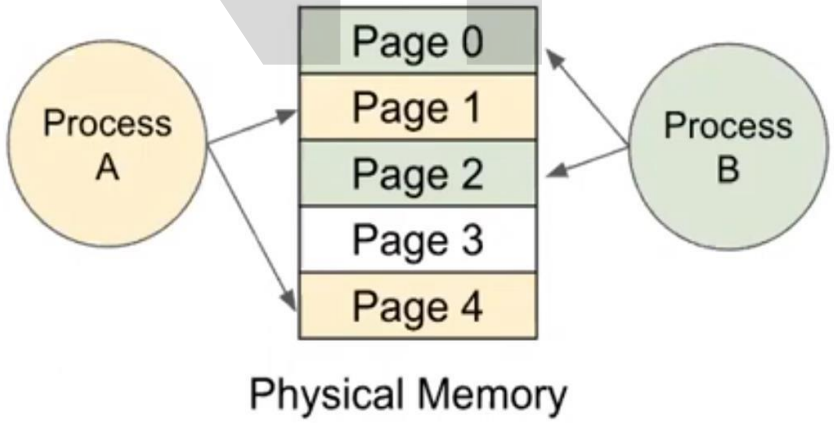
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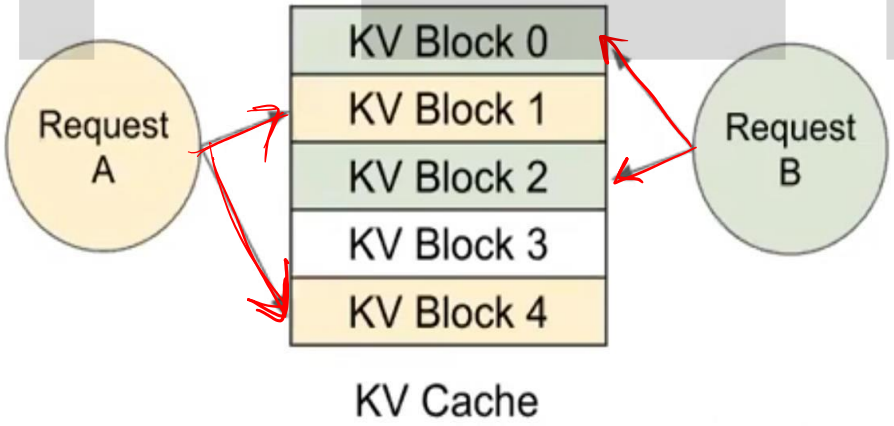
– Allocate on demand as a request grows

Inspited by OS **virtual memory** and **paging**!

Memory management in OS



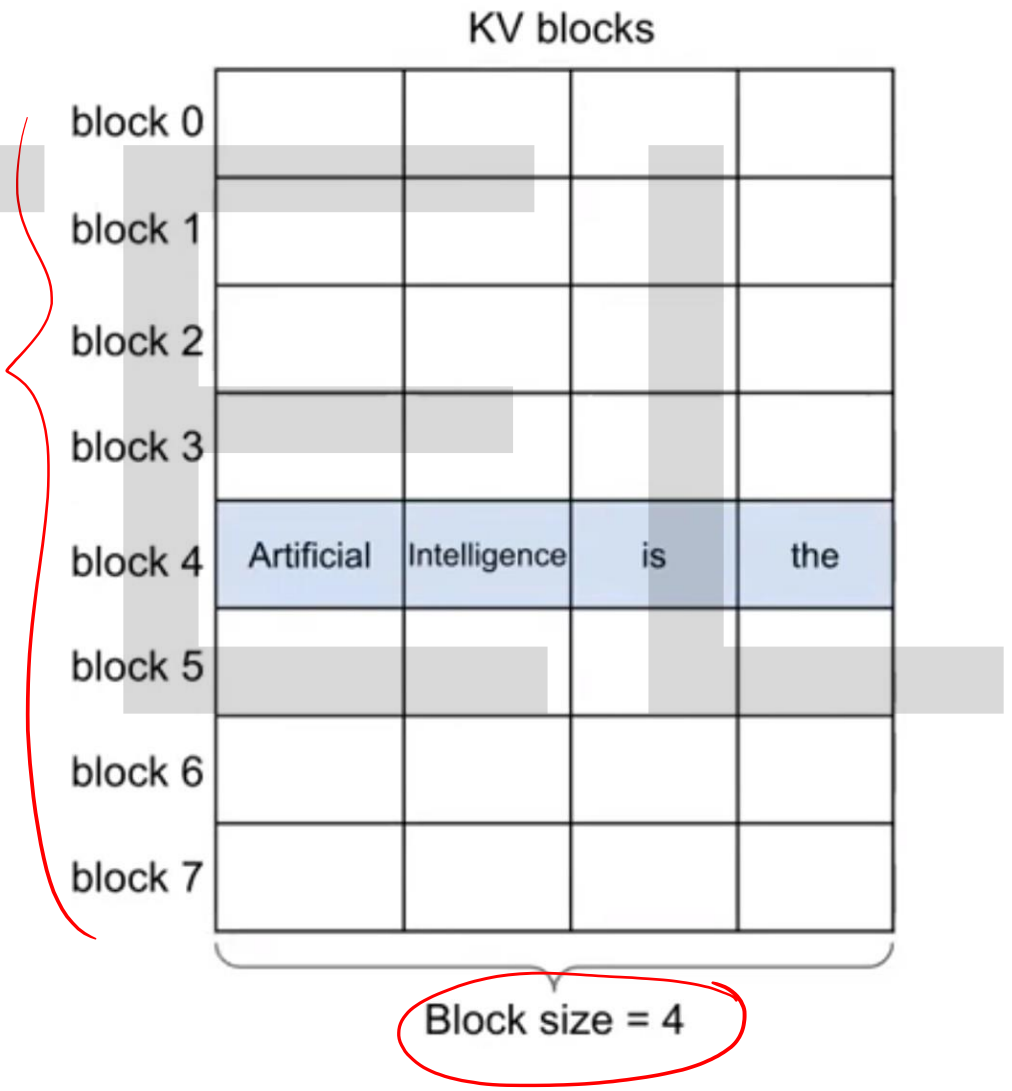
Memory management in vLLM



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KV Block

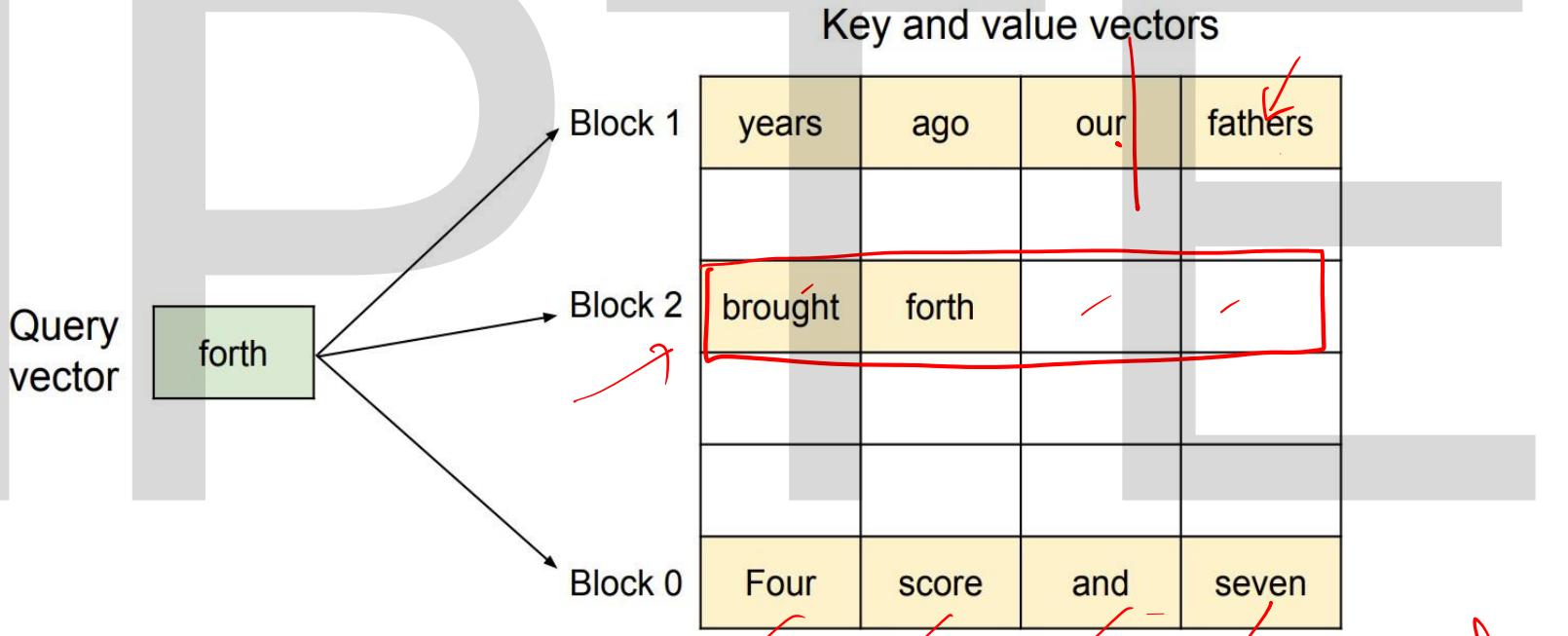
- A fixed-size contiguous chunk of memory that stores the tokens' KV states
- Similar to the concept of a memory page



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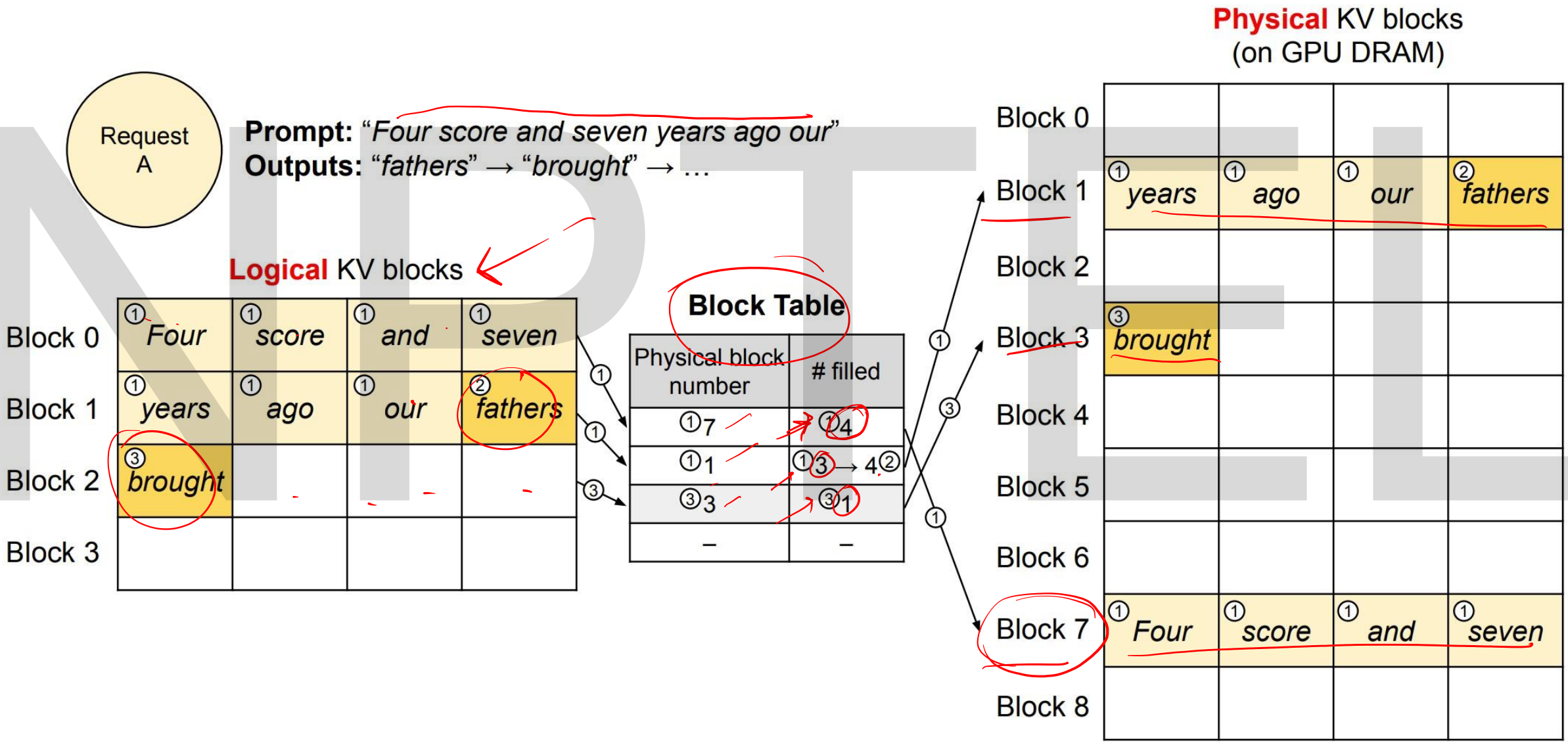
Paged Attention

- Manages KV cache in block granularity instead of sequence (i.e., request) granularity
- Allows storing logically contiguous KV blocks in non-contiguous physical memory

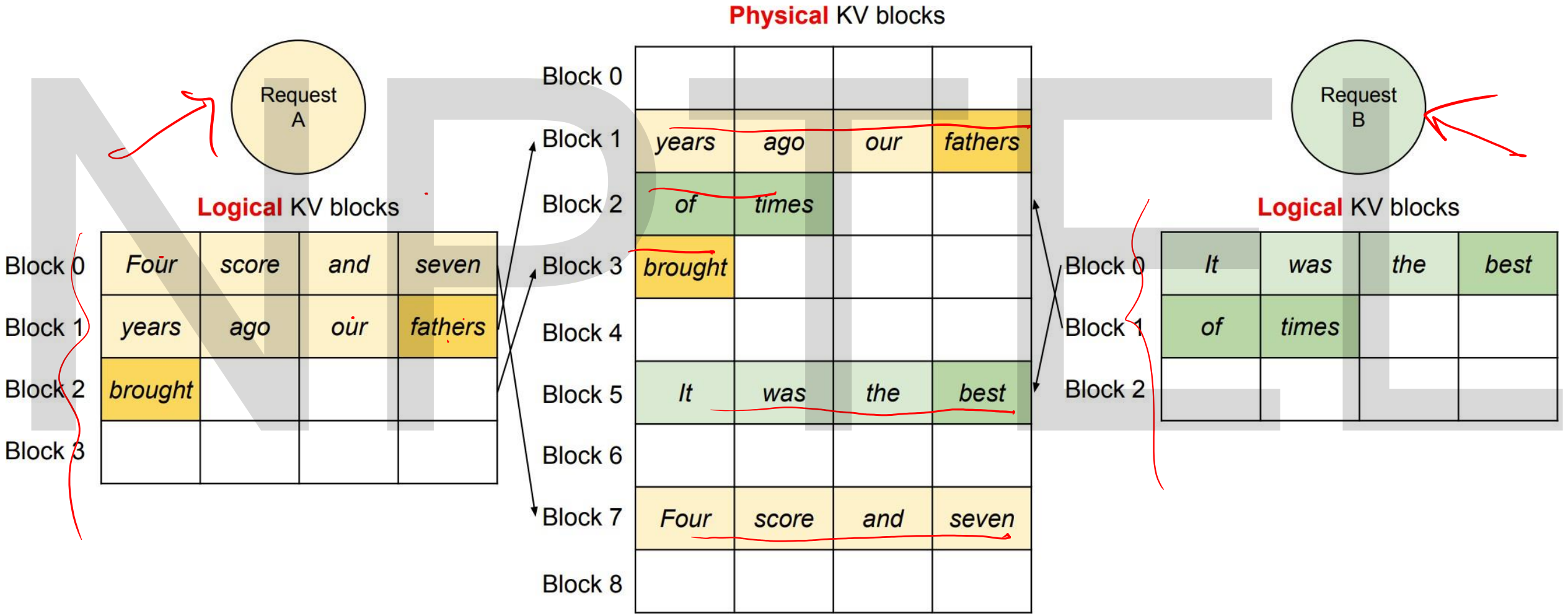


Example sequence: "Four score and seven years ago our | fathers brought forth"

Logical-to-Physical KV Block Translation



Serving Multiple Requests



Memory efficiency of vLLM

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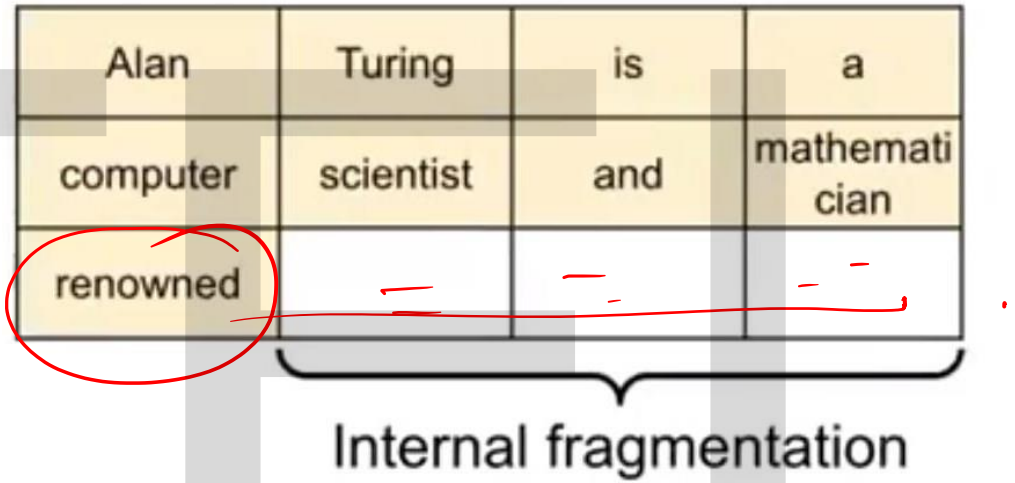
✓ No external fragmentation

Content credits: https://www.youtube.com/watch?v=5ZlavKF_68Ust=1646ssab_channel=Anyscale

LLMs: Introduction and Recent Advances

Memory efficiency of vLLM

- ✓ Minimal internal fragmentation
  - Only happens at the last block of a sequence
  - **# wasted tokens / seq < block size**
    - ➢ Sequence: O(100) or O(1000) tokens
    - ➢ Block size: 16 or 32 tokens
- ✓ No external fragmentation

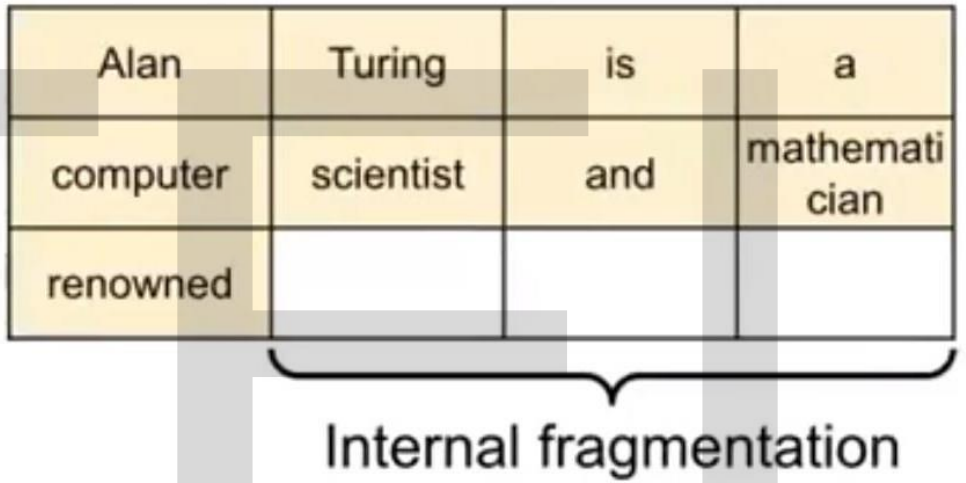


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LLMs: Introduction and Recent Advances

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- ✓ No external fragmentation
- ✓ On average, wasted space < **4%** of KV cache

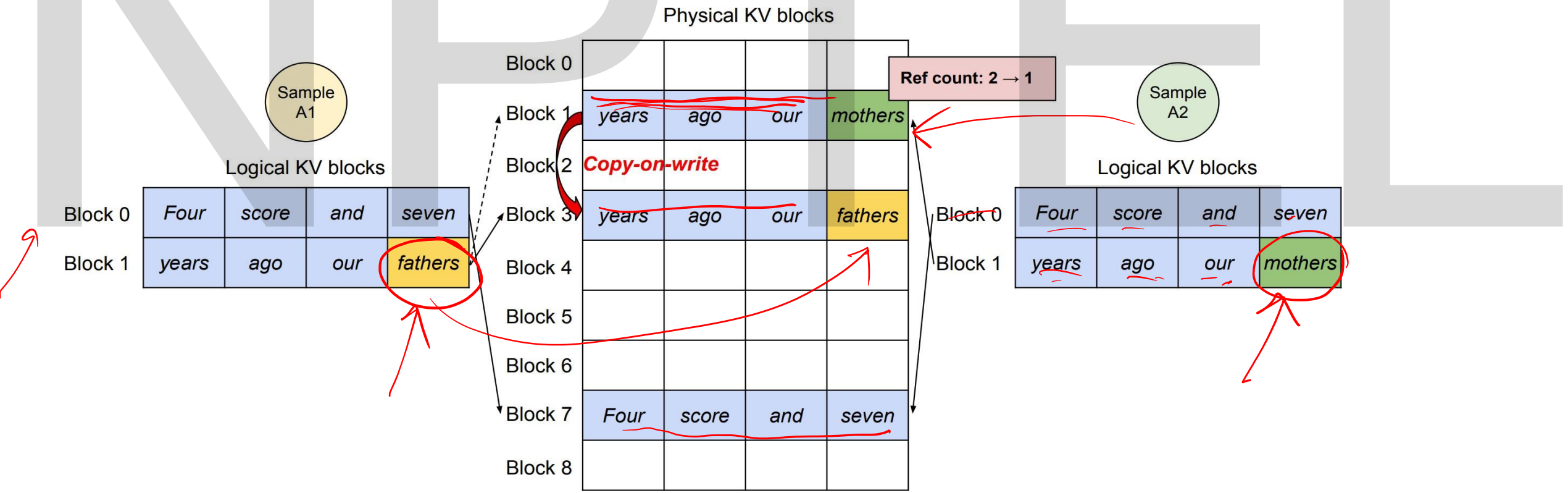


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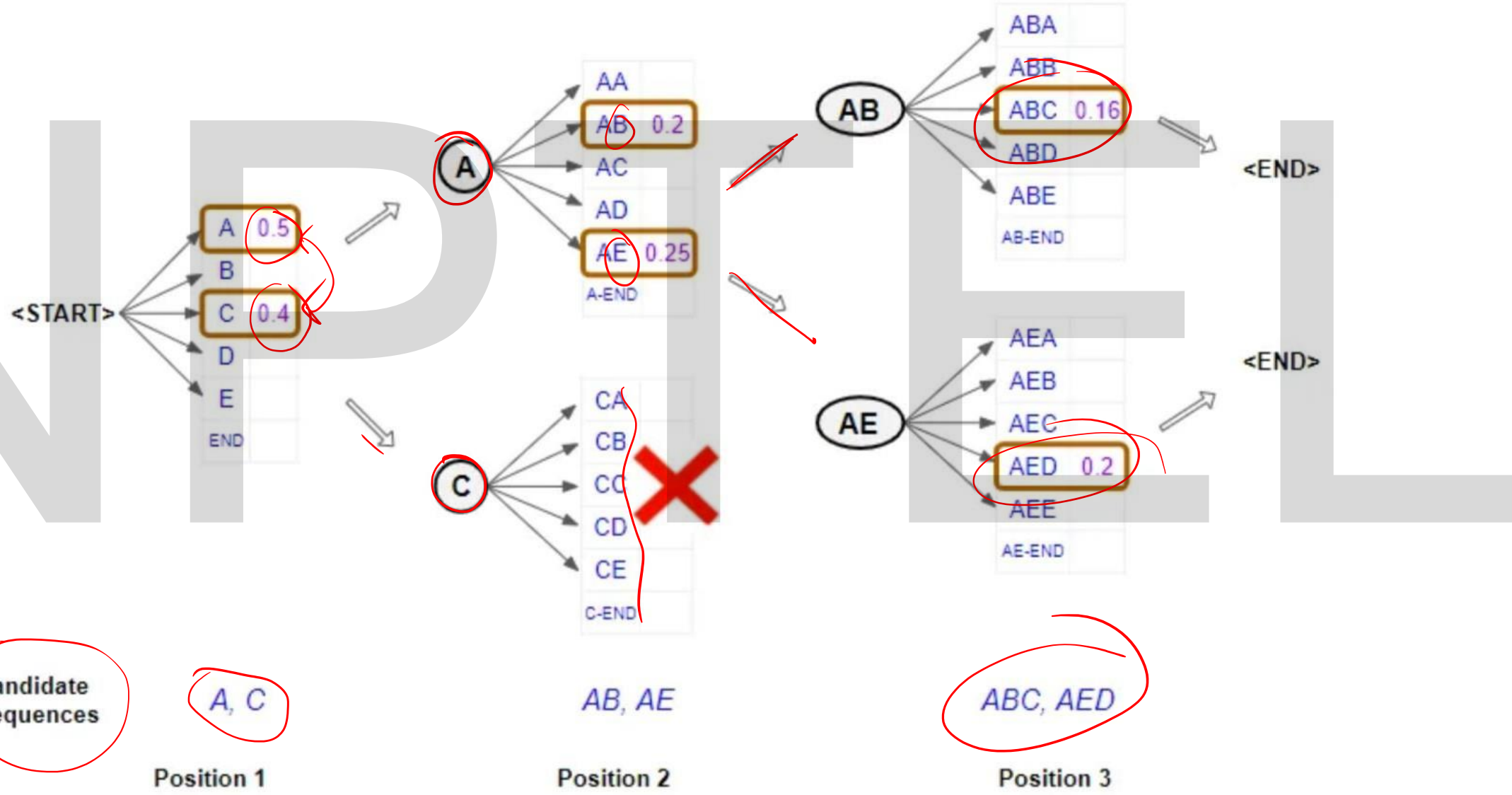
LLMs: Introduction and Recent Advances

1. Parallel Sampling

- **LLM generates multiple sampled outputs for a single input prompt; users can choose a favorite output from various candidates.**

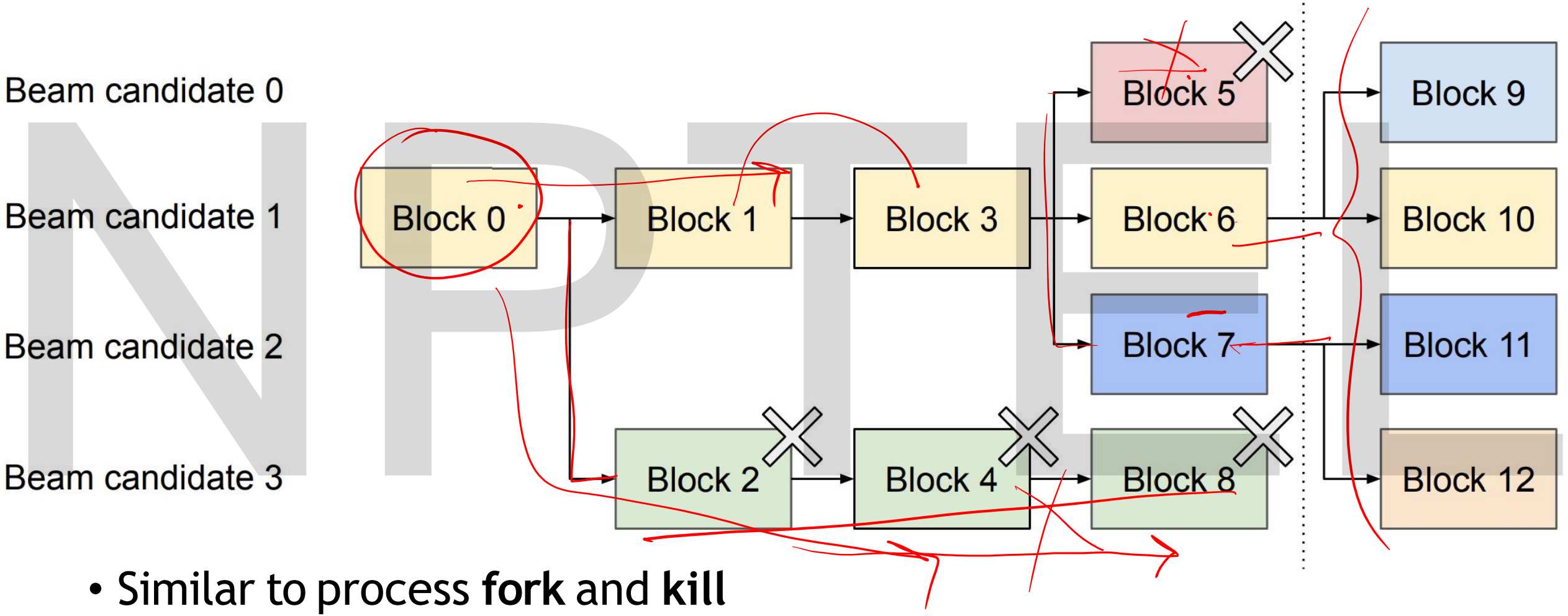


2. Beam Search



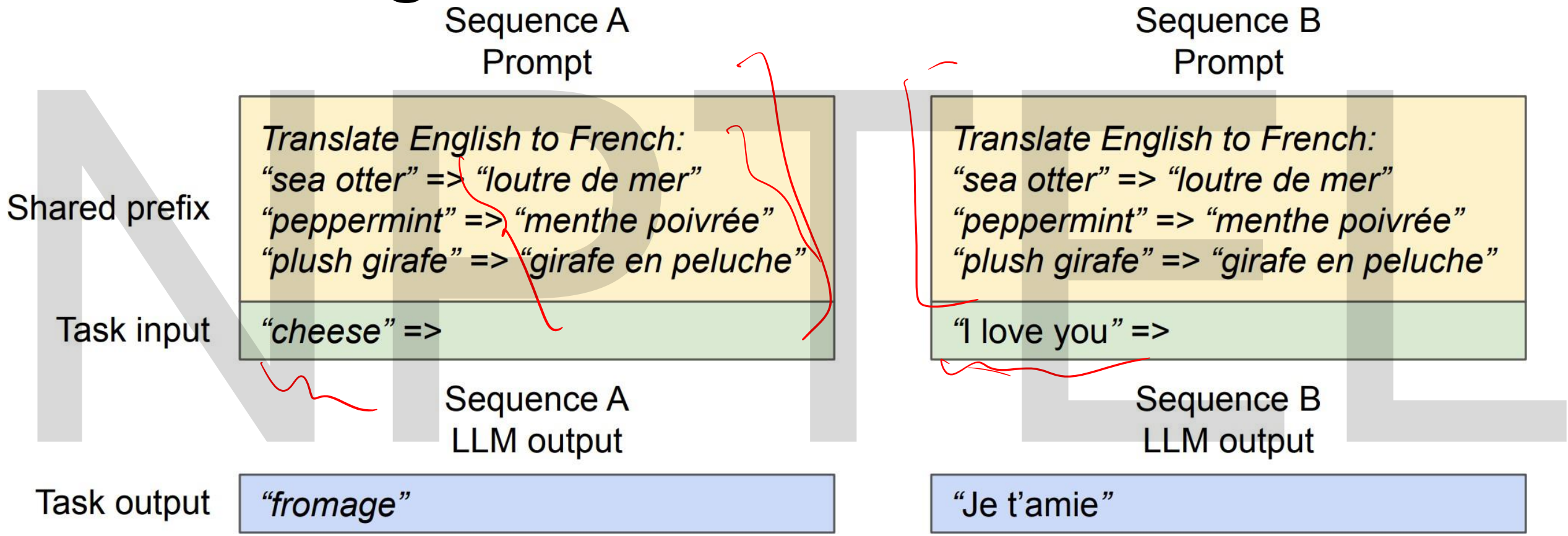
Picture source: https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24.

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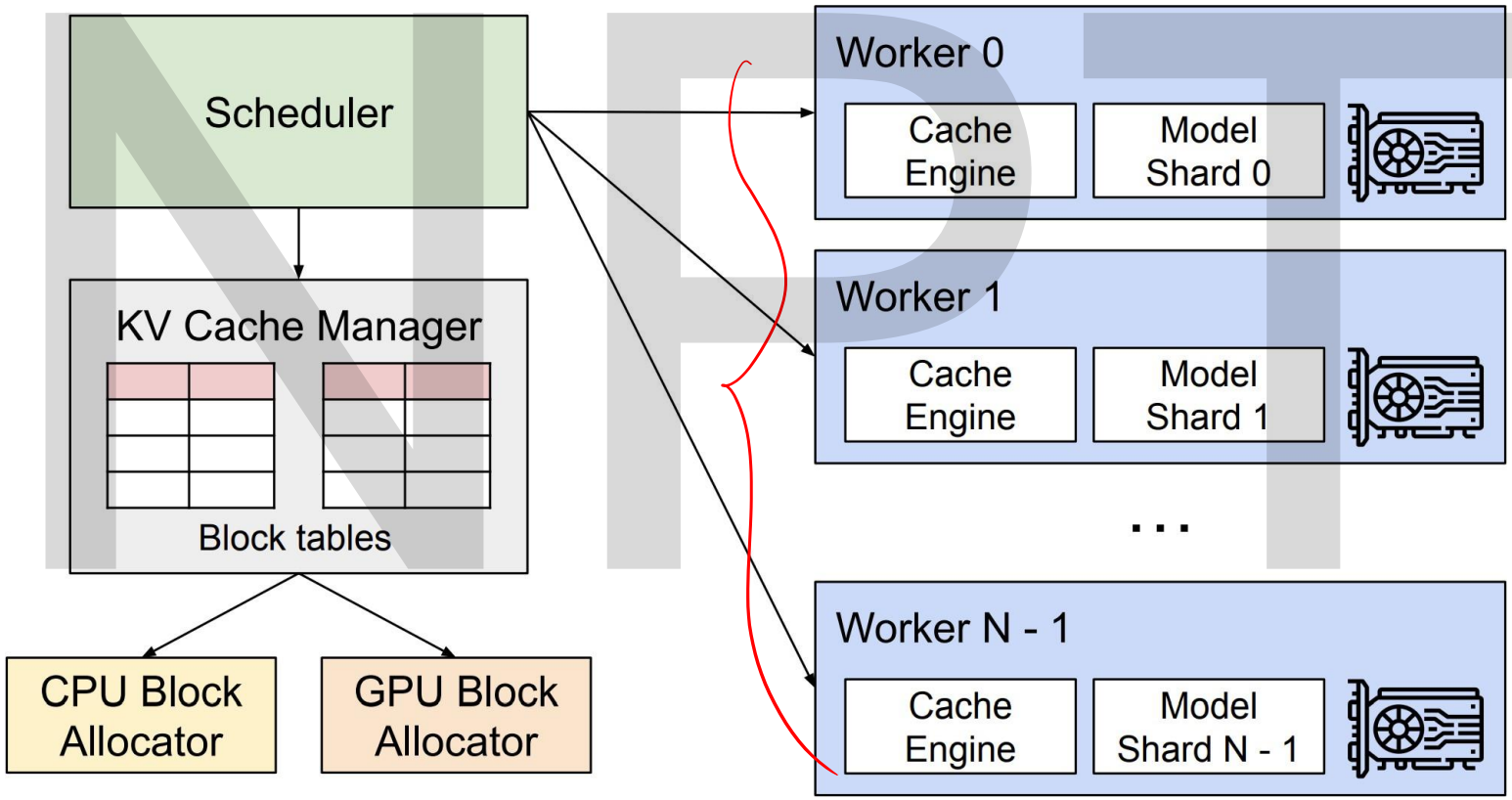
- Similar to process **fork** and **kill**

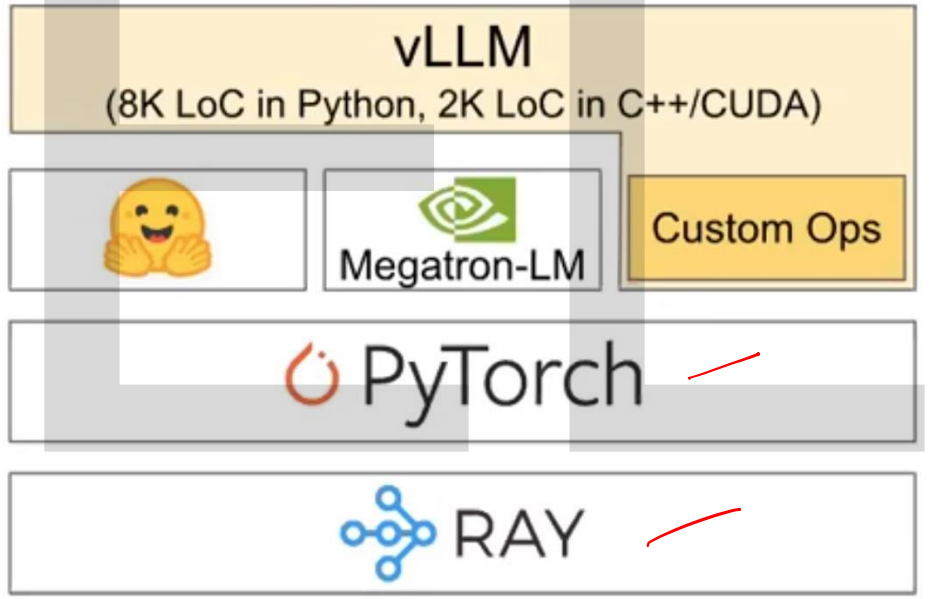
3. Prefix sharing



- Similar to shared libraries in OS, reserve few physical blocks for shared prefixes

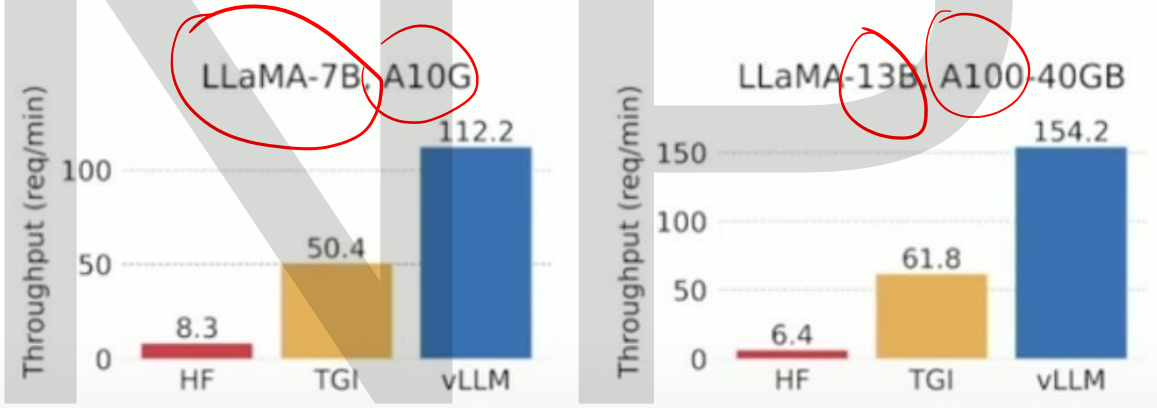
vLLM System Architecture Implementation



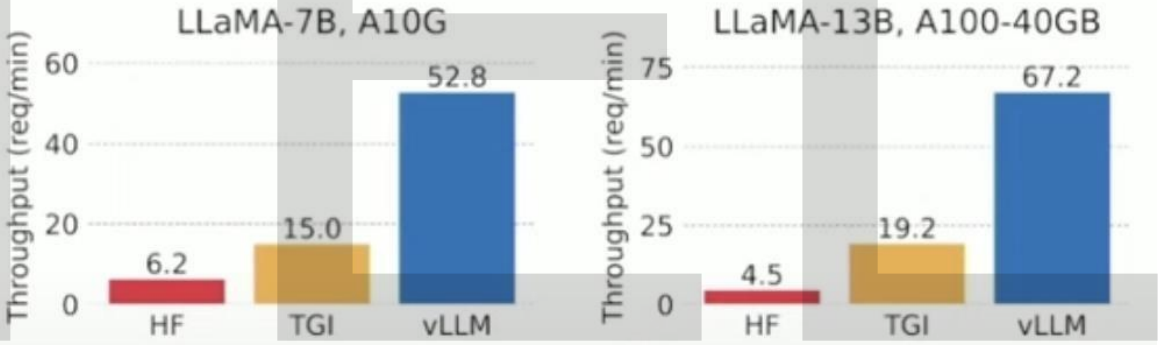


Comparison with HuggingFace and TGI (2023)

- Up to **24x** higher throughput than HuggingFace (HF)
- Up to **3.5x** higher throughput than Text Generation Inference (TGI)



Serving throughput when each request asks for 1 output completion.



Serving throughput when each request asks for 3 output completions.

vLLM Improves Inference Throughput by Enabling Larger Batch Size

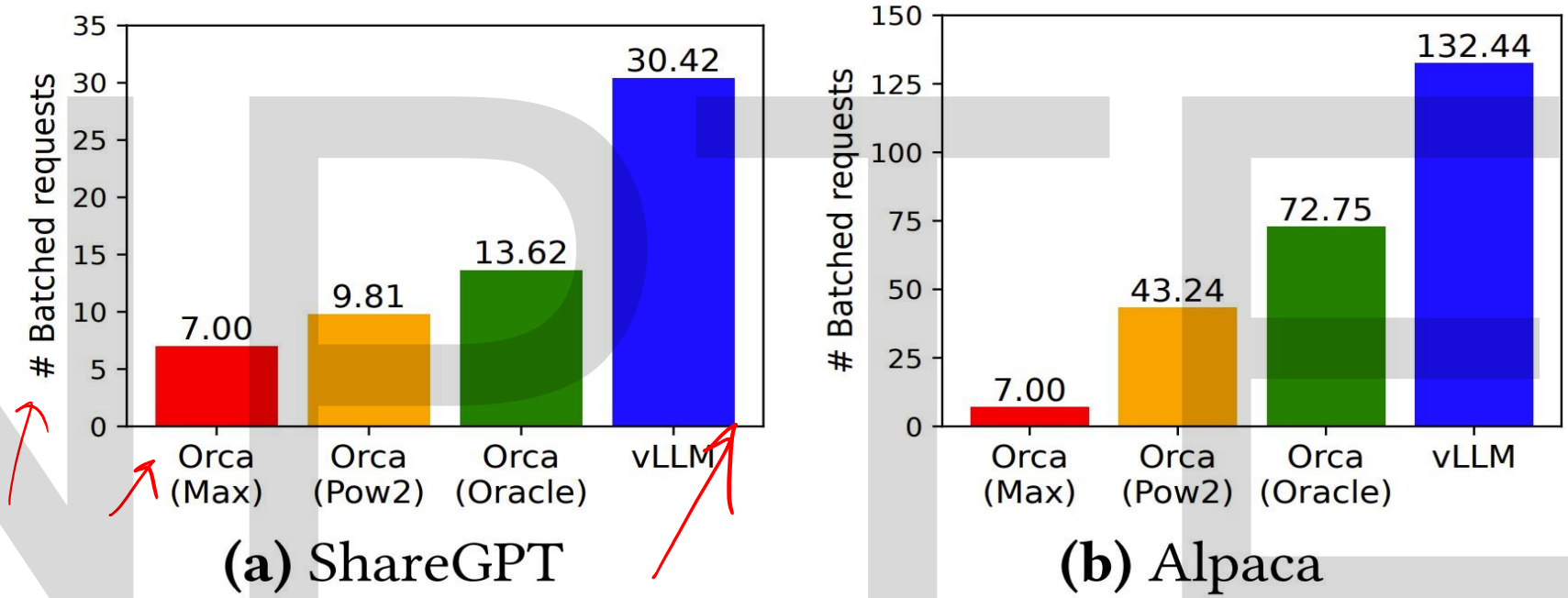
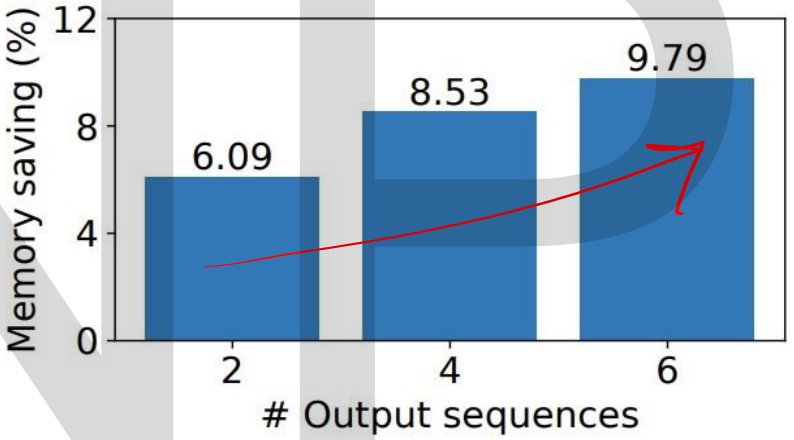
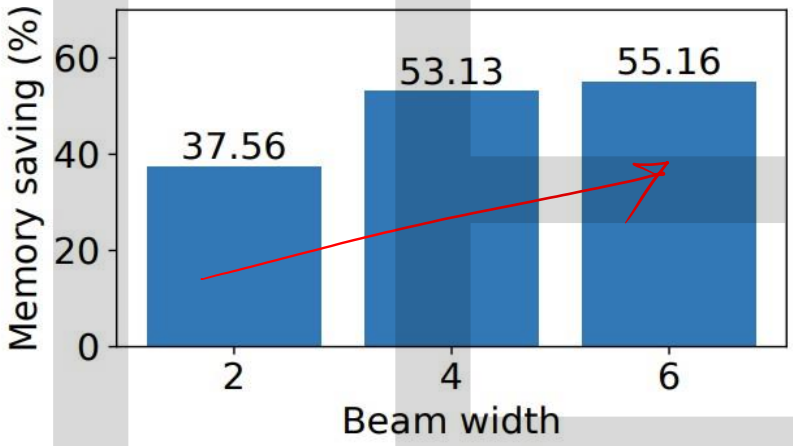


Figure 13. Average number of batched requests when serving OPT-13B for the ShareGPT (2 reqs/s) and Alpaca (30 reqs/s) traces.

Memory Saving of vLLM



(a) Parallel sampling



(b) Beam search

Figure 15. Average amount of memory saving from sharing KV blocks, when serving OPT-13B for the Alpaca trace.



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)